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Procedia Engineering 72 (2014) 823 – 828

Procedia Engineering

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The 2014 conference of the International Sports Engineering Association

Rebounding strategies in basketball

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Abstract

A three dimensional dynamic model is used to estimate the best rebounding position for players in basketball. The model has six distinct sub-models: flight, and ball-contact sub-models of ball-rim, ball-bridge, ball-board, ball-bridge-board, and ball-rim-board contact. It allows both slipping and non-slipping interactions at the ball-contact point. Using probabilistic selection of release velocity, angle, and lateral deviation angle, simulations analyse ball trajectories and good rebounding positions for players. Our estimation includes player wingspan. We analyse the relationship between the most likely player rebound positions and ball rebound positions with high probability. The best player rebound position is usually different from the most likely ball rebound position. Player rebound regions for players with large wingspans are very close to the hoop and include some likely ball rebound positions. For short-range shots, one player is able to collect almost all of rebounding balls but two or three rebounders are required for long-range shots.

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Selection and peer-review under responsibility of the Centre for Sports Engineering Research, Sheffield Hallam University

Keywords: basketball; player rebounding position; dynamic model

1. Introduction

Good basketball rebounders usually have height, the strength to box out, a sense of positioning, or leaping ability including a sense of timing. Some excellent NBA rebounders with about 7-ft wingspans collect more than 10 rebounds per game. They are able to cover two or three rebounding positions. Strong teams make use of the skill of good rebounders and formulate effective strategies to obtain rebounding balls.

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Selection and peer-review under responsibility of the Centre for Sports Engineering Research, Sheffield Hallam University
doi:10.1016/j.proeng.2014.06.140

Some previous studies have provided recommendations for good rebounding positions with observations or measurements in actual basketball games and even based on experiments (Allsen, 1967; Bryant, 1967; Huberty, 1970; Oba, 2009). Ribas et al. (2011) investigated the number of rebounders using measurements by two experienced observers. Another more deliberate and controlled method to estimate rebound position uses a simulation model. Okubo and Hubbard (2011; 2012; 2013) have calculated likely rebound positions and focused on the behaviour of the ball and where it goes in a probabilistic sense after unsuccessful shots. The related problem of where a rebounding player of a given arm length should be positioned to collect a maximum number of rebounding balls is different but related one.

To our knowledge, no previous study has analysed any rebounding strategy including player wingspan using a simulation model. We use a dynamic model and random Gaussian probability density functions based on previously measured shot percentages and standard deviations of release velocity and angle to calculate high-probability rebounding positions for players. We design a best player rebounding strategy to maximize collection probability.

2. Method

2.1. Dynamic model

Our dynamic model has six distinct sub-models: gravitational flight with air drag, and ball-contact sub-models for ball-rim, ball-backboard, ball-bridge (hoop-mounting plate), ball-bridge-board and ball-rim-board contacts. It switches between the sub-models depending on contact velocity and forces. The flight sub-model is adopted when the normal force at each contact point vanishes. Each contact sub-model has possible slipping and non-slipping, and spinning and non-spinning motions. The equations of motion have been derived for the flight and ball-contact sub-models including experimentally determined equivalent ball contact stiffness and damping.

2.2. Release conditions and assumption

A Newtonian frame with origin at the hoop centre has its *XY* plane horizontal, the direction *Y* from hoop centre toward the backboard, and *Z* up. Release location is specified by horizontal and vertical distance, l and h, from the hoop centre, and floor angle, β, between the board surface and the vertical plane including the release point and the hoop centre. Shots have four more parameters: release velocity, v, backspin angular velocity, ω, release angle, α, between the velocity and the horizontal plane, and lateral deviation angle, δ, between the initial ball path plane and the vertical plane including the hoop centre and release point.

We choose direct and bank shots with short ($l = 2$ m, $h = 0.05$ m and $\omega = 1$ Hz), medium ($l = 4$ m, $h = 0.15$ m and $\omega = 2$ Hz) and long ($l = 7$ m, $h = 0.30$ m and $\omega = 3$ Hz) ranges. Successful shot percentages for short, medium, and long shots are assumed to be 0.5, 0.45, and 0.4, respectively. Direct shots can occur at six different floor angles: 0° (near baseline), 15°, 30°, 45°, 60° and 75° (near court centre line), and bank shots at the last five of these. All shots occur from the right side of the court.

Aiming at the centre of the capture region allows the largest total error margin for the release velocity, angle and lateral deviation angle. We assume that shooters choose the centre of capture as their optimal aiming point for the three directions v, δ, α. We also use values chosen randomly from independent Gaussian probability density functions characterized by standard deviations of release velocity, angle and lateral deviation angle. Toyoshima et al. (1981) measured capture rates, release velocity, release angle and their standard deviations for 2.425, 4.225, and 6.025 m shots. The shot percentages and the standard deviations for skilled players were 74.2%, 0.18 m/s and 2.73°, 66.7%, 0.11 m/s, and 1.04°, and 37.5%, 0.15 m/s and 0.51° for their three kinds of shots, respectively. Referring to these measured data, we apply them to our short-, medium-, and long-range direct and bank shots and estimate the standard deviation of lateral deviation angle. For example, the short-range direct shot with a floor angle of 45° is assumed to have a shot percentage of 75%, and standard deviations of release velocity and angle of 0.18 m/s and 2.7°, respectively, and the standard deviation of lateral deviation angle can be calculated to be 0.16°.

3. Numerical simulations

3.1. Rebound probability density

We calculate rebound probability density where the ball falls within segmental court areas of 0.09^2 m^2. Rebound position is defined at the moment when the height of the ball centre equals hoop height on the way down. Figure 1 shows the top view of basketball court with rebound probability density plotted for direct and bank shots with floor angles of 0° and 45°. The dashed line for direct shots shows the top view of the optimal shot path plane including the hoop centre. For bank shots, the dashed line would be the top view of the optimal shot path plane if a rigid ball with no friction bounced off a no-hoop backboard with the angle of incidence equal to the reflection angle. High percentage regions show locations of high probability density. We assume that centre, power forward and small forward players possibly collect rebounding balls in the circle with a radius of 1, 0.9, and 0.8 m, respectively. We distinguish between ball rebound positions and player rebound positions: ball rebound positions have high probability density where the ball rebounds and the best player rebound positions are where players should be positioned to collect the maximum number of rebounding balls within the rebound circle. We draw rebound circles for centres in short-range shots, centres and power forwards in medium-range shots, and centres, power and small forwards in long-range shots in Fig. 1.

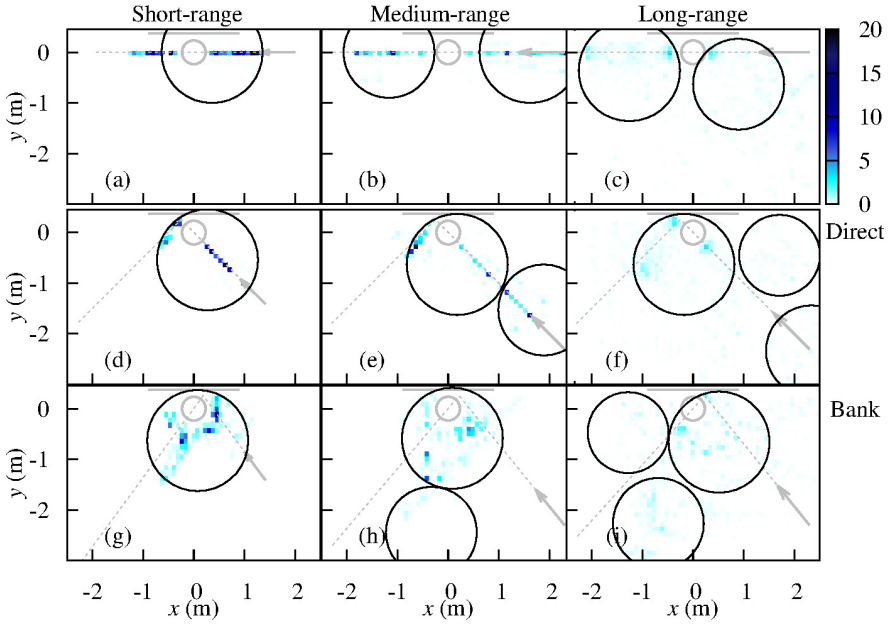


Fig. 1. Top view of basketball court with rebound probability density plotted for (a) short-, (b) medium-, and (c) long-range direct shots with a floor angle of 0°, for (d) short-, (e) medium-, and (f) long-range direct shots with a floor angle of 45°, and (g) short-, (h) medium-, and (i) long-range bank shots with a floor angle of 45°. Circles denote best rebounding regions for centres, power forwards, and small forwards.

Each missed shot has high probability ball rebound regions on the same side as and on the opposite side from the shooter. Angled direct and bank shots have different ball rebounding positions. Most likely ball rebound regions are in or near the optimal shot path plane. For short-range shots, centres apparently collect almost all rebounding balls. For long-range shots, two or three rebounders are required to collect rebounding balls. High probability ball rebound positions are usually different from the high probability player rebound positions. Centres are able to cover some high-probability ball rebound positions. Most likely player rebound positions for centres are near the hoop and in or near the optimal shot path plane.

Figure 2 shows rebound probability for centres, power forwards, and small forwards as a function of floor angle for short-, medium-, and long-range direct and bank shots. Rebound regions for centres usually have high probability. For short-range shots, centres are able to collect almost all rebounding balls. This may be because the ball does not bounce far away from the hoop and centres have long wingspans. For medium-range direct shots, two rebounders usually are needed to collect most rebounding balls. However, for medium-range bank shots with floor angles from 45° to 75°, centres have very high probabilities. In these shots, balls usually rebound near the hoop and centres are able to collect almost all rebounding balls. For long-range shots, player rebound probability for centres decreases and that for forwards increases. Ball rebound regions with high probability are not always close to one another for long-range shots. Thus, two or three rebounders are required. Ribas et al. (2011) have concluded that it is important to have superiority in number of rebounders. In our results, the number of rebounders roughly depends on the corresponding shot distance. One player is required for short-range shots and two or three players for long-range shots. Player rebound positions with high probability always exist and it is essential to cover them.

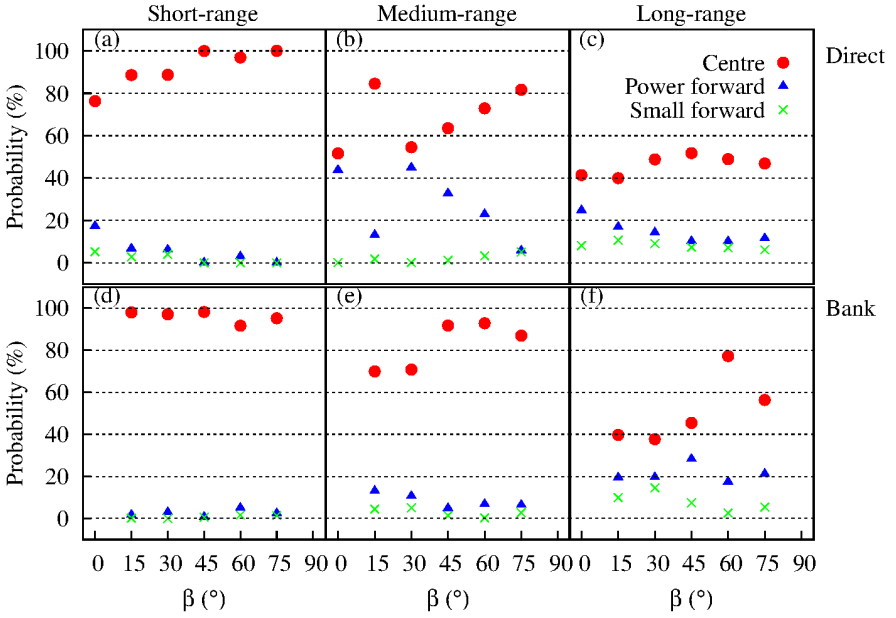


Fig. 2. Rebound probability in the most likely player rebound regions for centres, power forwards, and small forwards as a function of floor angle for (a) short-, (b) medium-, and (c) long-range direct, and (d) short-, (e) medium-, and (f) long-range bank shots.

3.2. Player rebounding positions

Figures 3 and 4 show rebound distance from the hoop and angle from the *X* axis for centres, power forwards and small forwards as a function of the floor angle for short-, medium-, and long-range direct and bank shots. Player rebound positions for short-range angled direct and bank shots are almost the same although Okubo and Hubbard (2013) have pointed out that angled direct and bank shots have different ball rebound positions. This may be because centres can collect rebounding balls in some ball rebound regions with high probability. For medium-, and long-range shots, angled direct and bank shots apparently have different player rebound positions.

Player rebound distance for centres usually is close to the hoop for short-, medium-, and long-range direct and bank shots. Especially, for short-range shots, player rebound distance is very close to the hoop. NBA Hall of fame forward Larry Bird said, “Most rebounds are taken below the rim. That’s where I get mine.” His remarkable position skill for rebounds may have compensated for his lack of leaping ability. The rebound distance for forwards is larger than that for centres in many types of medium- and long-range missed shots.

Previous studies have discussed which side of the hoop has an advantage for collecting rebounding balls. Some pointed out that the opposite side from the shooter has an advantage for rebound position, but others concluded that there was no difference between the two sides. Okubo and Hubbard (2013) have pointed out that it is difficult to find any comparative advantage for ball rebound positions when the player aiming points in the three release conditions are optimal, because balls may equally likely rebound to both sides. However, for some missed shots there is an advantage of either the side near the shooter or the opposite side for rebounds. For long-range direct shots, the best player rebound positions for centres are on the opposite side from the shooter, and those of power and small forwards are on the same side as the shooter. Best player rebound positions for centres are on the same side as the shooter for medium-range direct shots. For medium-range direct shots with floor angles from 0° to 30°, and from 45° to 75°, best player rebound positions for power forwards are on the opposite side from and on the same side as the shooter, respectively. For many missed bank shots, best player rebound positions for centres are in front of the hoop. For medium-range bank shots with floor angles larger than 45°, power forwards should be positioned on the opposite side from the shooter.

Instruction of both the most likely ball rebound position and best player rebound regions may be very useful for rebounders. Players will be positioned at the most likely player rebound position and prepare for rebounding balls at the maximum likelihood ball rebound position.

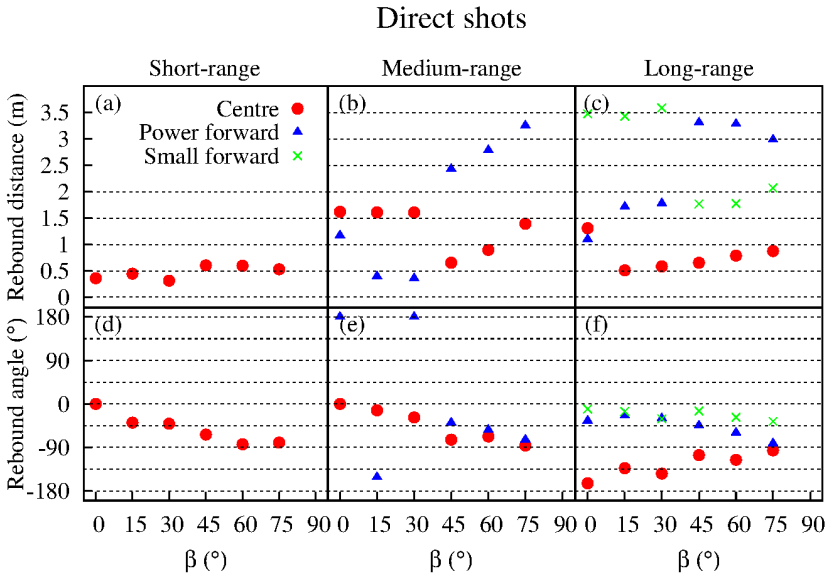


Fig. 3. Best rebound distance from the hoop for centres, power forwards, and small forwards (a) short-, (b)medium-, and (c) long-range direct shots; and angle from the *X* axis for (d) short-, (e) medium-, and (f) long-range direct shots, all as a function of floor angle.

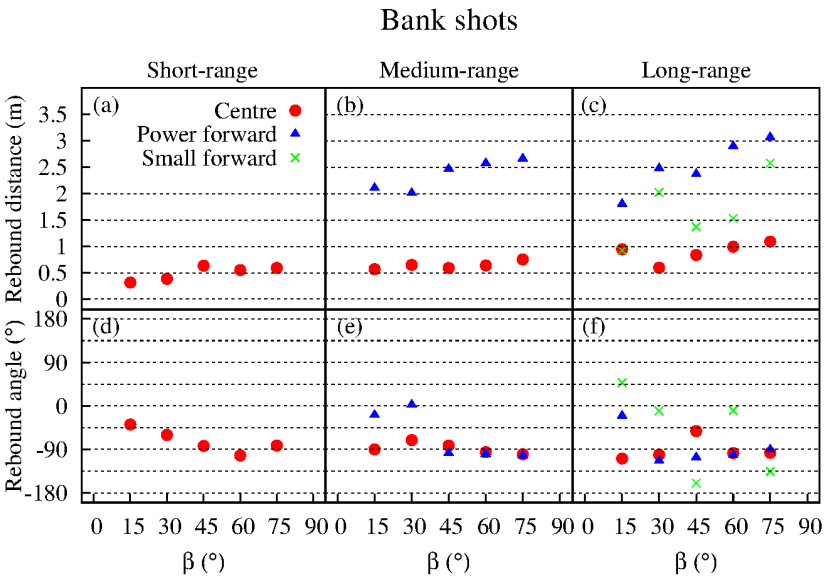


Fig. 4. Rebound distance from the hoop for centres, power forwards, and small forwards (a) short-, (b) medium-, and (c) long-range bank shots; and angle from the *X* axis for (d) short-, (e) medium-, and (f) long-range bank shots, all as a function of floor angle.

4. Conclusions

We have calculated best rebounding positions for rebounders with different wingspans after short-, medium-, and long-range missed direct and bank shots using a dynamic model and random Gaussian probability density functions based on measured release conditions. We also have analysed the relationship between player rebound positions with high probability and likelihood ball rebound positions. The best player rebound regions usually are different from the most likely ball rebound positions and include some very likely ball rebound positions. The number of rebounders needed roughly depends on the corresponding shot distance: one player for short-range shots and two or three for long-range shots. Players with long wingspans should be positioned close to the hoop for rebounds because balls rebound near the hoop in many cases and they can collect rebounding balls in a large area. These results may help players understand the most likely ball and player rebound positions. They establish the best positions for rebounds and allow players to prepare for predicted rebounding balls at the most likely positions.

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